

Section 10 Integration Comparison with Snowflake

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Agenda

- RDS & DynamoDB
- Spark
- Glue
- Iceberg
- Quicksight
- Athena
- Snowflake Comparison

Integration – DynamoDB

- What is DynamoDB? NoSQL key-value store.
 - OLTP database
- Move from OLTP to OLAP/DWH.
- Zero-ETL
 - Zero-ETL integration
 - Incremental data changes are moved every 15-30 mins.
- COPY command (similar to S3)

```
COPY <redshift-table> FROM 'dynamodb://<dynamodb-table>'  
IAM_ROLE 'arn:aws:iam::0123456789012:role/<redshift-role>';
```

Integration – RDS & Aurora

- Zero-ETL

Integration – Spark

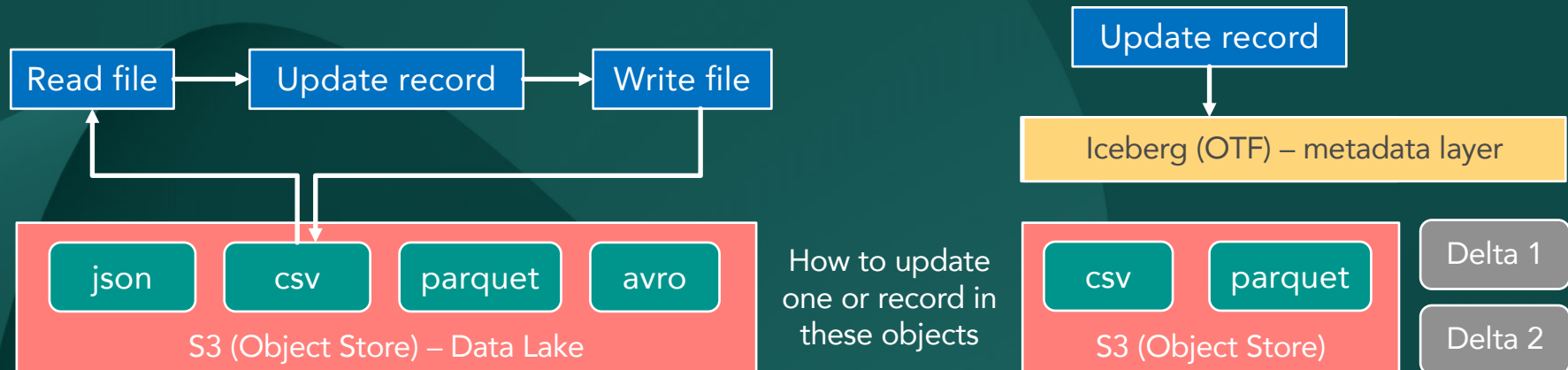
- What is Spark?
 - Distributed processing framework for batch and stream processing.
 - Used in the "T" of "ETL". Transformation of data.
 - For more details refer to "Data Engineering - Vol 2" in www.3aayaam.in
- Redshift Spark connector
 - Read data from Redshift using Spark
 - Write data to Redshift using Spark
- Integration is automatically included in EMR and Glue.

Integration – Glue

- Glue – Managed ETL that runs open source Spark.
- Use Redshift connections to move data in and out.

Integration – Iceberg

- What is Iceberg and Open Table Format (OTF).
- Record or row level modifications are difficult.
- OTF format (Iceberg, HUDI, Delta) solves this problem.
- Provides Lakehouse (Data Lake + Data Warehouse)

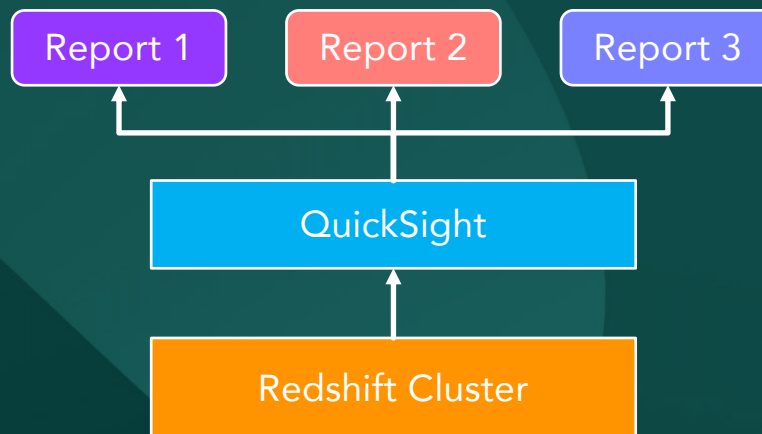


Integration – Iceberg

- Redshift Spectrum and Serverless can be used to query Iceberg tables.
- Redshift does not support writing to Iceberg tables. READ-ONLY.
- Athena and Spark on EMR handles writes.
- MVs can be created on Iceberg tables just like any other external tables on S3.
- Lake Formation can be used for fine-grained access control.

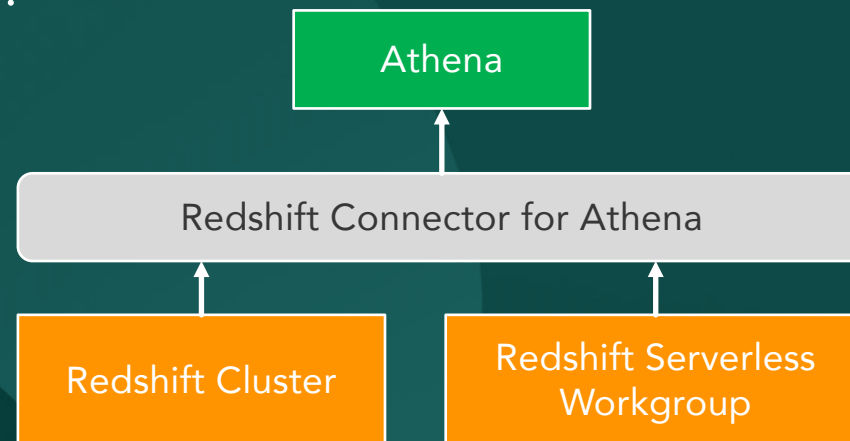
Integration – QuickSight

- What is QuickSight?
- BI, reporting and visualization tool on AWS. Like PowerBI, Tableau, Qlik, Looker etc.
- Consumer for Redshift.



Integration – Athena

- What is Athena?
- Interactive query capability (using SQL) on top of Data Lake.
- Can run queries on Redshift. Only reads.
- Demo will be in Athena chapter.



Snowflake Comparison

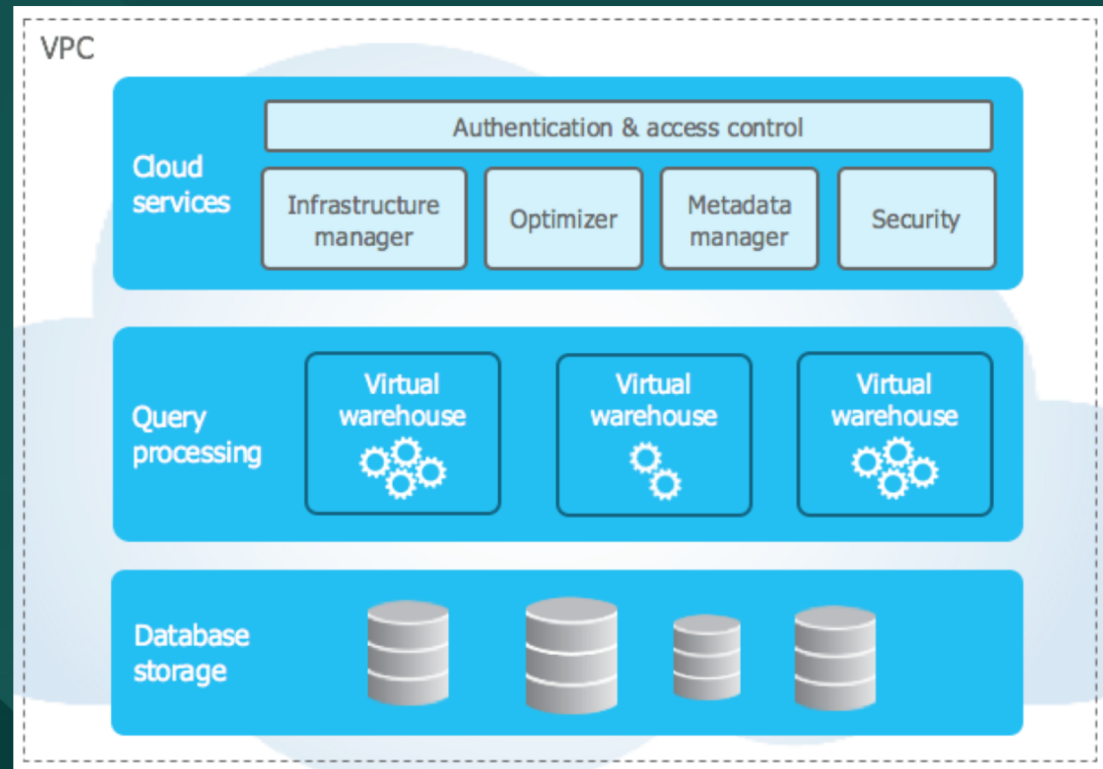
- Technical comparison
- Architectural nuances
- Feature differences

What is Snowflake

- What is Snowflake?
- Data Platform – storing, processing and analyzing data.
- Data Warehouse.
- Deployment can be on AWS, Azure or GCP.
- No software installation, configuration and management.
- Maintenance, management, upgrades are handled by Snowflake.

Snowflake Architecture

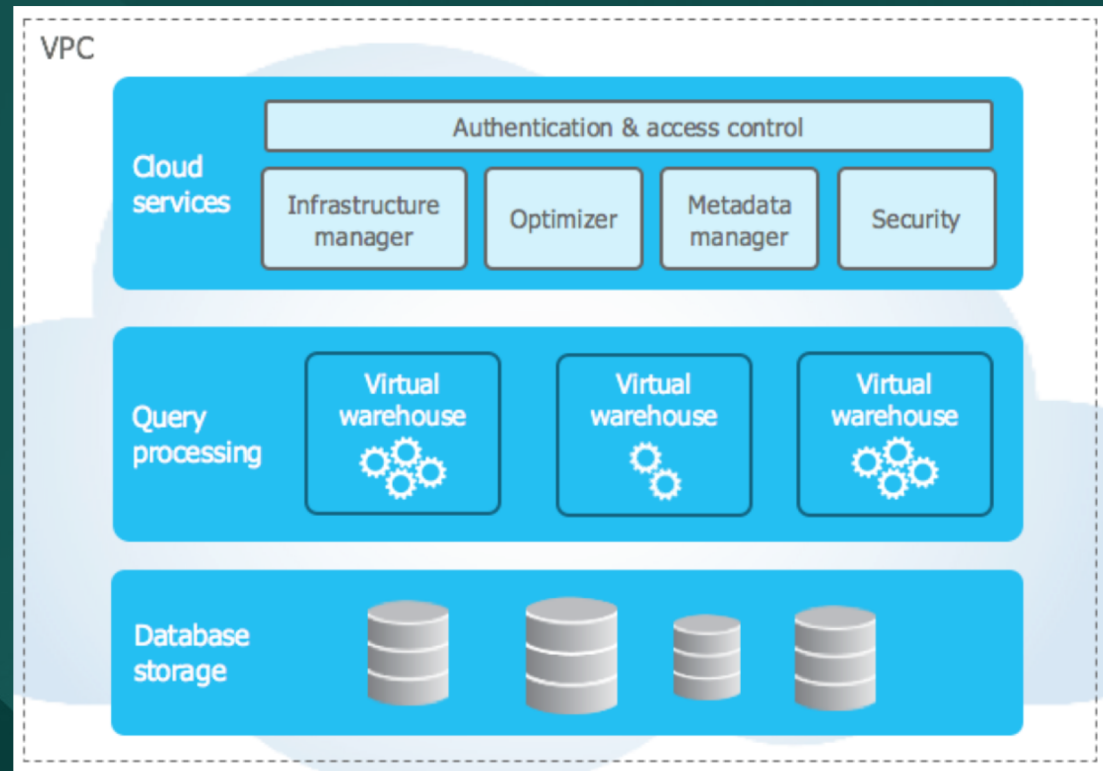
- Storage – Columnar.
 - Reorganization (Vacuum).
 - Compression.
 - Uses cloud storage (S3, GCS).
- Compute (VWH) – MPP Cluster.
 - Each node also holds portion of the data locally.
 - Virtual warehouse is an independent compute cluster.



Picture courtesy – docs.snowflake.com

Snowflake Architecture

- Cloud Services – Coordinate activities across Snowflake.
 - Collection of micro-services.
 - Authentication.
 - Authorization.
 - Optimization and execution plan.



Picture courtesy – docs.snowflake.com

Virtual Warehouse

- Cluster of Compute Resources.
 - CPU
 - RAM
 - Temporary Storage
 - Executing SQL statements
 - Pause and resume
 - No Leader Node
- Multi-cluster warehouse (auto-scaling, dynamic or static)

- Redshift Compute Nodes
 - CPU
 - RAM
 - Local Disk
 - Slices
 - Pause & Resume
 - SQL queries are handled by LN
- Cluster Resize
- Concurrency Scaling

Database Objects

- Snowflake Tables.
 - Micro-partitions & Data Clustering
 - Clustering Keys
 - Automatic Clustering
- External Tables (S3, GCS, Azure Blob)
- Iceberg Tables
 - Both Read and Write (only while using Snowflake Catalog)
- Views & Materialized Views (MR & AR)

- Redshift Tables
 - Zone Maps
 - Sort Keys
 - Automatic Table Optimization
- External Table (S3)
- Iceberg Tables using Spectrum
 - Only Reads
- Views & Materialized Views (MR & AR)

Data Movement

- Load and Extract data.
 - COPY INTO command (S3, GCS, Azure Blob)
 - Copies to a staging table first.
 - COPY INTO to unload data.
- Snowpipe - Loads data from file as soon as it is available in S3, GCS, Azure Blob.
- Snowpipe Streaming – Continuous ingestion into tables.
- Semi-structured data using VARIANT, OBJECT and ARRAY data types.
- Supports unstructured data (image, video, audio, text etc.) using cloud storage and internal storage of Snowflake.
- Dynamic tables

- Load and Extract Data
 - COPY command (S3, DynamoDB) – Direct load from S3 to Tables.
 - UNLOAD command.
- Continuous file ingestion (preview)
- Streaming ingestion using auto-refreshable MVs.
- SUPER data type.
- No direct support for unstructured data.
- AutoMV.

Snowflake Features

- Horizon Catalog – Component to discover, govern and secure data.
 - Discover data – Metadata Repository.
 - Security – Authentication, Authorization, CLS, RLS, encryption-at-rest and in transit.
 - Governance and compliance – Access history, data lineage
- Connecting to and accessing Snowflake data.
 - SnowSQL
 - Python, Spark
 - JDBC, ODBC
 - BI tools – QuickSight, Looker, PowerBI, MicroStrategy etc.

- AWS Glue Catalog
- Security –
 - IdP federation
 - IAM Authentication
 - Native database
- RLS, CLS, Encryption
- Governance and compliance
- Connecting to Redshift
 - RSQL
 - Python, Spark
 - JDBC, ODBS
 - BI Tools

Snowflake Features

- Snowsight – Web interface for Snowflake.
 - Execute SQL queries.
 - Monitor query performance and query history.
 - Write Snowpark Python.
 - Visualizations.
- Snowpark – Queries and data transformation using DataFrame like APIs of Spark and Pandas.

- Query Editor V2
 - Execute SQL queries
 - Query history
 - Notebook
 - Charts
 - Scheduled queries

Thank You

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